

Table 4: Accuracy & matching time on IwPIS dataset (K=1)

Algorithm	Accuracy (%)	Time (ms)
SC	40.78	3.14×10^0
SC+DP	37.59	1.40×10^{-1}
IDSC	30.92	2.92×10^0
IDSC+DP	34.90	1.36×10^{-1}
SRV	36.40	8.44×10^2
DBCSR (opt)	33.75	4.12×10^4
DBCSR (uni)	27.72	4.20×10^4
HSC	35.15	1.72×10^{-2}
MDM	27.06	4.08×10^{-2}
HF	36.96	1.42×10^1
Proposed MORT	44.79	1.56×10^{-2}

vNets method, DCL (?), and then investigate the comparative results of ConvNets methods and the fused MORT. To facilitate the comparison between the proposed method and ConvNets methods, we combine the fused MORT (as the feature encoder) with SVM (as the classifier) using the same protocol as adopted by ? (?).

Competitors. We compare the proposed MORT with the following state-of-the-art ConvNets methods: (1) Three state-of-the-art ConvNets models: Alexnet (?), VGG-16 (?), and ResNet-50 (?); and (2) three fine-grained state-of-the-art methods: improved B-CNN (?), fast-MPN-COV (?), and DCL (?).

Implementation details. All the models are implemented in Pytorch 1.0.0 and are pretrained on the ImageNet (?) and then fine-tuned on each dataset. In all experiments, the input images are resized to 440×440 , and cropped to 384×384 randomly for training. Standard data augmentations are applied including random rotation within ± 15 degree and horizontal flip with 0.5 probability. For fast-MPN-COV and improved B-CNN, the models are trained with the best setting in the implementation of fast-MPN-COV (?). For the remaining methods, the models are trained with the default settings in the implementation of DCL (?).

Evaluation on patchy image structure datasets. For SoyCultivarVein, BtfPIS and IwPIS datasets, we select the first half images from each category as the training set, and the remaining images as the testing set. Table 5 lists the classification accuracies of all the competing ConvNets methods on the three datasets. The fused MORT achieves the best classification accuracy of 65% and 98.00% on SoyCultivarVein dataset and BtfPIS dataset, respectively. Nevertheless, the fused MORT ranks third in classification accuracy on IwPIS, with 6% and 12% lower than the improved B-CNN and fast-MPN-COV, respectively. A possible reason is that the fused MORT has much lower feature dimension (1×2092) per image compared with improved B-CNN (1×262144) and fast-MPN-COV (1×32896). The overall results show that the fused MORT surpasses the original MORT in classification accuracy on all the three patchy

Table 5: Accuracy on three patchy image structure datasets

Algorithm	Accuracy (%)		
	SoyCultivarVein	BtfPIS	IwPIS
Alexnet	14	93.20	44
VGG-16	16	95.60	48
ResNet-50	28	96.80	52
improved B-CNN	63	73.60	74
fast-MPN-COV	51	77.20	80
DCL	39	97.20	68
fused MORT	65	98.00	68

image structure datasets, indicating that ConvNets methods and the proposed MORT can provide each other with complementary information. The superior results of the fused MORT over ConvNets methods on three datasets indicate that the proposed MORT can extract discriminative features even with very limited training samples, leading to the performance improvements of fused MORT over the sole use of DCL or original MORT.

Conclusion

In this paper, we presented a novel Multi-Orientation Region Transform (MORT) method, which is rotation, translation and scale invariant, for effective and efficient classification of patchy image structures. The proposed MORT can extract local structural features at various scales and orientations for comprehensive shape description. The encouraging experimental results on three patchy image structure databases demonstrate the effectiveness and efficiency of MORT for patchy image shape classification. Moreover, the results also indicate that the proposed MORT and ConvNets methods can provide each other with important complementary features to further improve the classification performance.

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